

Solution Requirements

OptiCrop — Smart Agricultural Production Optimization Engine

Functional Requirements

- FR1: Accept N, P, K, temperature, humidity, pH, and rainfall as numeric inputs from the user.
- FR2: Return the most suitable crop recommendation based on the entered conditions.
- FR3: Provide an About page describing the project's purpose and methodology.
- FR4: Support training and comparison of multiple ML algorithms (Logistic Regression, KNN, Decision Tree, Random Forest, K-Means) so the best-performing one can be selected for deployment.
- FR5: Persist the trained model so predictions can be served without retraining on every request.

Non-Functional Requirements

- NFR1: Prediction response time under ~2 seconds.
- NFR2: Usable by someone with no ML background — plain form inputs, plain-language output.
- NFR3: Target model accuracy of at least 90% on held-out test data.
- NFR4: Architecture should be extensible to new datasets, regions, or algorithms without a full redesign.